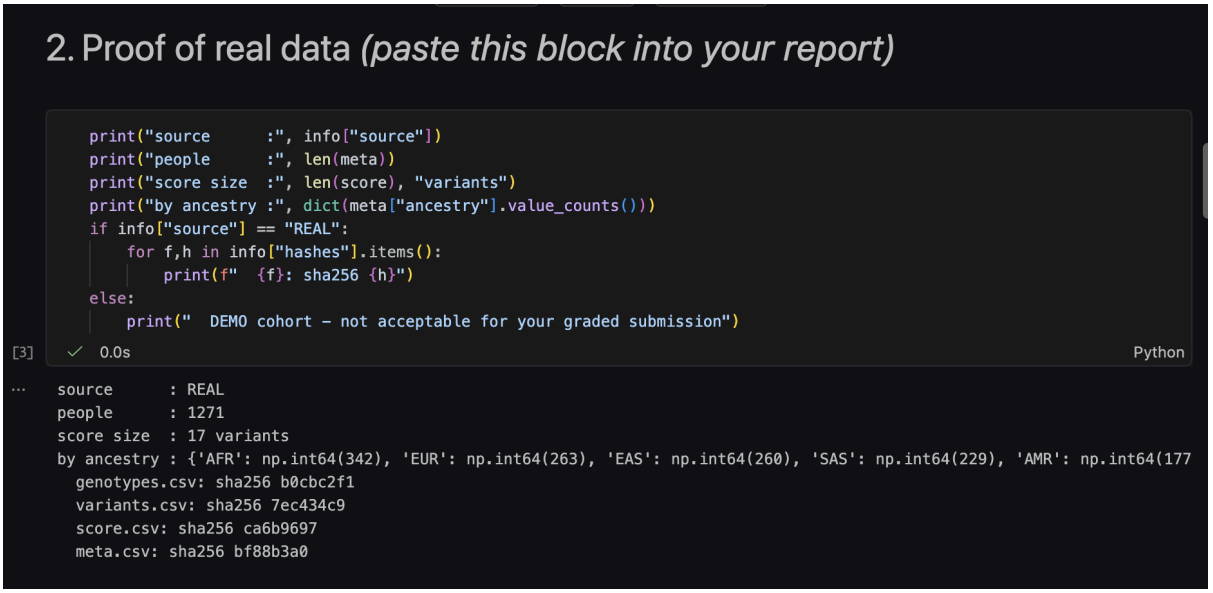
Module 6 Assignment - New



This picture shows that the source is real.

The genotype data comes from the third phase of the 1000 Genomes Project, using GRCh37 as the genome version. The polygenic risk score (PRS) of ovarian cancer comes from PGS000048 in the PGS Catalog database.

The final data set contains 1,271 female samples and 17 genetic variations associated with the risk of ovarian cancer. By checking the positions and alleles of genetic mutations, all 17 mutations were successfully matched to the genotype data.

Ancestry information was from the 1000 Genomes sample metadata. Age and ovarian cancer phenotype labels were simulated because the dataset does not contain actual ovarian cancer diagnoses data. Therefore, the AUC and calibration results in this study only reflect the performance of the model on simulated disease labels rather than on real ovarian cancer data.

1.Why does your European-derived score lose accuracy in other ancestries? Point to LD and allele-frequency differences, not biology.

The ovarian cancer polygenic risk score (PGS000048) was established using the genetic association data of the European ancestral population. So, when it is applied to other ancestral groups, the prediction accuracy may decrease.

The one main reason is differences in linkage disequilibrium, also called LD. A certain SNP in PRS may be highly correlated with the variation that really affects the disease in the European ancestral group, but in other ancestral groups, this correlation may be weaker.

Additionally, the frequency of alleles between different ancestral groups may also be different, affecting the distribution and predictive performance of PRS.
In the analysis, the simulated AUC was highest in EUR, which is 0.727, while the AUC of SAS was lowest, which is 0.650. And the AUC of AFR is 0.695, the AMR is 0.688 and the EAS is 0.681. However, the disease label of ovarian cancer is partly based on the simulation of PRS itself. Therefore, these AUC differences do not prove that PRS has poor predictive ability in patients with real non-European ancestral ovarian cancer, nor can it prove that the difference is caused by LD.

2. One threshold, many groups. Using your 3.3 numbers, is a single top-10% cutoff fair? What happens if a clinic applies it unchanged?
Using the top 10% threshold for all ancestries may cause fairness problems because there are differences in the distribution of PRS in different populations. Although this cutoff divides about 10% of individuals in the entire study cohort into high PRS groups, the proportion of people exceeding the threshold within different ancestral groups may be very different.
In our ovarian cancer analysis, from 3.3 in the notebook, there is 31.2% population in the EUR above the cutoff, while there is only 0.6% above the cutoff in the AFR groups. The proportions of EAS, SAS and AMR are 3.5%, 8.3% and 7.9% respectively. These differences show that the same threshold may cause some ancestries to be more frequently classified as high risk.
One possible reason is that the frequency of alleles in different groups of people is different, which leads to a shift in the distribution of PRS. In addition, differences in linkage disequilibrium, also called LD, may also affect the ability of genetic markers to reflect related disease mutations in different ancestral groups.
If a clinic directly uses this cutoff without verifying it further, the patients from EUR are easier to get additional screening and genetic research, while other groups' patients may be ignored, which will cause the fairness problem in the clinical opportunity.

3. Discrimination vs calibration. What did each tell you (3.4, 3.5), and where did they disagree?
Discrimination and calibration evaluate two different aspects of PRS performance. Discrimination measures how well the PRS separates individuals with and without the

disease, usually using AUC. Calibration measures whether predicted disease probabilities agree with observed disease frequencies.

From 3.4 and 3.5 in the notebook, the total AUC is 0.701. The EUR's AUC is highest, which is 0.727, followed by AFR that is 0.695, and then is 0.688 in AMR and 0.681 in EAS, and finally SAS is the lowest, which is 0.650. These results show that in our simulation data, there are differences in the distinguishing ability of PRS in different ancestral groups.

From the 3.5 in the notebook, we use EUR samples to establish the risk prediction model and then apply it in all the ancestry groups. The calibration curves showed that there are some deviations with the ideal diagonal line, which means the predicted probabilities are not always same with the real proportions of simulated cases.

The discrimination and the calibration may come out different results, because although a model can arrange the risk of individuals correctly, it may also overestimate or underestimate the probability of disease. Therefore, AUC is higher doesn't mean the calibration must be better. Although EUR's AUC is the highest, 0.727, it cannot be concluded that the prediction probability of EUR is completely accurate. Therefore, both AUC and calibration curves are needed to evaluate PRS performance.

4. Clinical utility and limits. Where could the score help, where should it not be used, and what would you need before deploying it? Note that the phenotype is simulated, and name the real data you would need to validate for real.

The PRS can be a useful additional tool for identifying groups that may require further risk assessment or preventive attention. From 3.6 in the notebook, the AUC age only shows 0.567 and it improves to 0.717 after adding the PRS, which means the discrimination improves with PRS.

However, the PRS should not be used alone for disease diagnosis, nor should it be directly used as the only evidence for making clinical decisions. The notebook 3.4 and 3.5 shows differences in discrimination and calibration in ancestry groups, which means PRS may not provide reliable predictions for each group. So, we cannot judge whether a person is sick or needs treatment based on his PRS score alone.

Since the phenotype in this notebook is simulated, real disease outcomes are needed before clinical use. Before deploying the score, we need to use the real data of patients to validate

the model, including real disease diagnoses, genetic information, patient information and so on. We also need to test the discrimination and calibration of the model in other populations.